

Math 121: Commutative Algebra

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Worksheet 2.2: Algebraic Sets

Theorem 0.1 (Hilbert’s Nullstellensatz I). *If $\mathbb{K}$ is algebraically closed, then every maximal ideal of $\mathbb{K}[x_1, \dots, x_n]$ is of the form $\langle x_1 - a_1, \dots, x_n - a_n \rangle$ for some point $(a_1, \dots, a_n) \in \mathbb{K}^n$.*

Note this is just the first of a series of theorems that will go by Hilbert’s Nullstellensatz. Recall that a $\mathbb{K}$ -algebra is a ring R together with a specified ring homomorphism $\mathbb{K} \rightarrow R$. A morphism of $\mathbb{K}$ -algebras $\varphi : R \rightarrow S$ is a ring homomorphism that preserves scalars, i.e. $\varphi(c \cdot 1_R) = c \cdot 1_S$ for every $c \in \mathbb{K}$. In particular, $\mathbb{K}[x_1, \dots, x_n]$ is a $\mathbb{K}$ -algebra, every evaluation map $\text{ev}_p : \mathbb{K}[x_1, \dots, x_n] \rightarrow \mathbb{K}$ is a $\mathbb{K}$ -algebra homomorphism, and if $\mathfrak{m}$ is a maximal ideal then $R/\mathfrak{m}$ is naturally a field equipped with a $\mathbb{K}$ -algebra structure. When we say below that a residue field is “isomorphic to $\mathbb{K}$,” we mean *as a $\mathbb{K}$ -algebra*.

Problem 1. Evaluation Maps. For a point $p = (a_1, \dots, a_n) \in \mathbb{K}^n$ the *evaluation-at- p* map is given by

$$\begin{array}{ccc} \mathbb{K}[x_1, \dots, x_n] & \xrightarrow{\text{ev}_p} & \mathbb{K} \\ f & \longmapsto & f(p) \end{array} .$$

- (a) Prove that ev_p is a ring homomorphism and find generators for $\mathfrak{m}_p := \ker(\text{ev}_p)$.
- (b) Prove that there is a bijection, induced by these evaluation maps, between $\mathbb{K}^n$ and the subset of $\text{Spec}(\mathbb{K}[x_1, \dots, x_n])$ consisting of maximal ideals whose residue field is isomorphic to $\mathbb{K}$ as a $\mathbb{K}$ -algebra.
- (c) Find a maximal ideal of $\mathbb{R}[x, y]$ that is not of the form $\mathfrak{m}_p$ for any $p \in \mathbb{R}^2$.
- (d) Restate Hilbert’s Nullstellensatz I (above) in terms of $\mathfrak{m}_p$.

Solution.

- (a) To show that ev_p is a ring homomorphism, we need to show that for all $f_1, f_2 \in \mathbb{K}[x_1, \dots, x_n]$ the three conditions hold,

Additivity:

$$\text{ev}_p(f_1 + f_2) = (f_1 + f_2)(p) = f_1(p) + f_2(p) = \text{ev}_p(f_1) + \text{ev}_p(f_2)$$

Multiplicativity:

$$\text{ev}_p(f_1 \cdot f_2) = (f_1 \cdot f_2)(p) = f_1(p) \cdot f_2(p) = \text{ev}_p(f_1) \cdot \text{ev}_p(f_2)$$

And $\text{ev}_p(1) = 1$ since 1 is a constant function.

(b) **TODO:**

(c) **TODO:**

(d) **TODO:**

Problem 2. Algebraic Sets. Let $\mathcal{S} \subset \mathbb{K}[x_1, \dots, x_n]$ be a subset. Define the corresponding *algebraic set* in $\mathbb{K}^n$ to be:

$$\mathbb{V}(\mathcal{S}) := \{p \in \mathbb{K}^n \mid f(p) = 0 \text{ for all } f \in \mathcal{S}\}.$$

(a) Prove that $\mathbb{V}$ is order-reverse, i.e., if $\mathcal{S} \subset \mathcal{T}$ then $\mathbb{V}(\mathcal{T}) \subset \mathbb{V}(\mathcal{S})$.

(b) Prove that $\mathbb{V}(\mathcal{S}) = \mathbb{V}(\langle \mathcal{S} \rangle) = \mathbb{V}(\sqrt{\langle \mathcal{S} \rangle})$.

(c) Describe each of the following algebraic sets, and when possible draw a picture.

(i) $\mathcal{S} := \{x^2 - y\} \subset \mathbb{R}[x, y]$.

(ii) $\mathcal{S} := \{x^2 - xy\} \subset \mathbb{R}[x, y]$.

(iii) $\mathcal{S} := \{x^6 - 3x^5y + 3x^4y^2 - x^3y^3\} \subset \mathbb{R}[x, y]$.

(iv) $\mathcal{S} := \{x^4 - 2x^3y + x^2y^2\} \subset \mathbb{R}[x, y]$.

(v) $\mathcal{S} := \{x^2 + y^2 - 1\} \subset \mathbb{R}[x, y]$.

(vi) $\mathcal{S} := \{n(x^2 + y^2 - z^2) \mid n \in \mathbb{Z}_{\geq 0}\} \subset \mathbb{R}[x, y, z]$.

(vii) $\mathcal{S} := \{x^2 + 1\} \subset \mathbb{R}[x]$.

(viii) $\mathcal{S} := \{x^2 + 1\} \subset \mathbb{C}[x]$.

(ix) $\mathcal{S}$ is a collection of any linear polynomials in $\mathbb{K}[x_1, \dots, x_n]$.

(d) Explain why $\text{SL}_n(\mathbb{K})$ has the structure of an algebraic set.

(e) Explain why the set of $m \times n$ matrices of rank $< t$ is an algebraic set contained in $\mathbb{K}^{m \times n}$.

Solution.

(a) Assume $\mathcal{S} \subset \mathcal{T}$. Let $\mathbf{p} \in \mathbb{V}(\mathcal{T})$.

Then, for all $f \in \mathcal{T}$, $f(\mathbf{p}) = 0$. But since $\mathcal{S} \subset \mathcal{T}$ this implies $g(\mathbf{p}) = 0, \forall g \in \mathcal{S}$. Hence, $\mathbf{p} \in \mathbb{V}(\mathcal{S})$. Since $\mathbf{p}$ was an arbitrary point in $\mathbb{V}(\mathcal{T})$ we conclude that $\mathbb{V}$ is order reversing.

(b) By definition,

$$\mathcal{S} \subseteq \langle \mathcal{S} \rangle \subseteq \sqrt{\langle \mathcal{S} \rangle}.$$

And from part (a) we have,

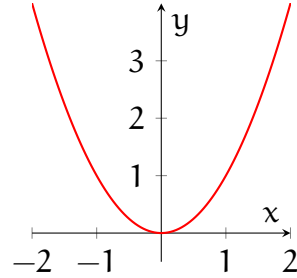
$$\mathbb{V}(\mathcal{S}) \supseteq \mathbb{V}(\langle \mathcal{S} \rangle) \supseteq \mathbb{V}(\sqrt{\langle \mathcal{S} \rangle}).$$

To show $\mathbb{V}(\langle \mathcal{S} \rangle) \subseteq \mathbb{V}(\sqrt{\langle \mathcal{S} \rangle})$ we let $\mathbf{p} \in \mathbb{V}(\langle \mathcal{S} \rangle)$. That is for all $f \in \langle \mathcal{S} \rangle \subseteq \sqrt{\langle \mathcal{S} \rangle}$, $f(\mathbf{p}) = 0$, i.e. $\mathbf{p} \in \mathbb{V}(\sqrt{\langle \mathcal{S} \rangle}) \implies \mathbb{V}(\langle \mathcal{S} \rangle) \subseteq \mathbb{V}(\sqrt{\langle \mathcal{S} \rangle})$.

To show that $\mathbb{V}(\mathcal{S}) \subseteq \mathbb{V}(\langle \mathcal{S} \rangle)$ we let $\mathbf{p} \in \mathbb{V}(\mathcal{S})$. That is for all $f \in \mathcal{S}$, $f(\mathbf{p}) = 0$ but since $\langle \mathcal{S} \rangle$ is the set of linear combinations of functions that evaluate to zero at $\mathbf{p}$, they also evaluate to zero at $\mathbf{p}$ and $\mathbf{p} \in \mathbb{V}(\langle \mathcal{S} \rangle)$.

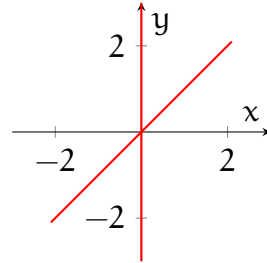
(c) (i) $\mathcal{S} := \{x^2 - y\} \subset \mathbb{R}[x, y]$.

$$\mathbb{V}(\mathcal{S}) = \{(x, y) \in \mathbb{R}^2 \mid x^2 = y\}$$



(ii) $\mathcal{S} := \{x^2 - xy\} \subset \mathbb{R}[x, y]$.

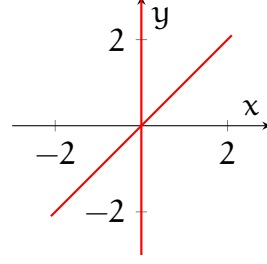
$$\mathbb{V}(\mathcal{S}) = \{(x, y) \in \mathbb{R}^2 \mid x = 0, x = y\}$$



(iii) $\mathcal{S} := \{x^6 - 3x^5y + 3x^4y^2 - x^3y^3\} \subset \mathbb{R}[x, y]$.

Note that $x^6 - 3x^5y + 3x^4y^2 - x^3y^3 = x^3(x - y)^3$.

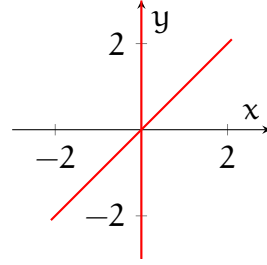
$$\mathbb{V}(\mathcal{S}) = \{(x, y) \in \mathbb{R}^2 \mid x = 0, x = y\}$$



$$(iv) \mathcal{S} := \{x^4 - 2x^3y + x^2y^2\} \subset \mathbb{R}[x, y].$$

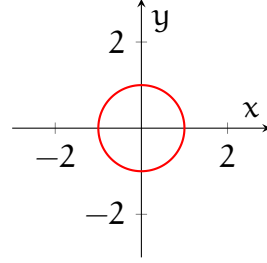
$$\text{Note that } x^4 - 2x^3y + x^2y^2 = x^2(x - y)^2.$$

$$\mathbb{V}(\mathcal{S}) = \{(x, y) \in \mathbb{R}^2 \mid x = 0, x = y\}$$



$$(v) \mathcal{S} := \{x^2 + y^2 - 1\} \subset \mathbb{R}[x, y].$$

$$\mathbb{V}(\mathcal{S}) = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$$



$$(vi) \mathcal{S} := \{n(x^2 + y^2 - z^2) \mid n \in \mathbb{Z}_{\geq 0}\} \subset \mathbb{R}[x, y, z].$$

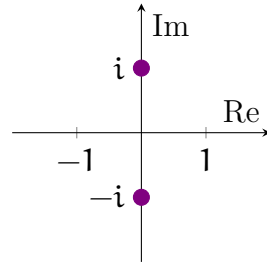
$$\mathbb{V}(\mathcal{S}) = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 = z^2\}$$

$$(vii) \mathcal{S} := \{x^2 + 1\} \subset \mathbb{R}[x].$$

$$\mathbb{V}(\mathcal{S}) = \emptyset \text{ since this equation has no solutions over } \mathbb{R}[x]$$

$$(viii) \mathcal{S} := \{x^2 + 1\} \subset \mathbb{C}[x].$$

$$\mathbb{V}(\mathcal{S}) = \{(x, y) \in \mathbb{R}^2 \mid x^2 = y\}$$



(ix) $\mathcal{S}$ is a collection of any linear polynomials in $\mathbb{K}[x_1, \dots, x_n]$.

$\mathbb{V}(\mathcal{S}) = \ker(A)$, where A is a $m \times n$ matrix where the i -th row is $(a_{i_1}, \dots, a_{i_n})$

(d) $\mathrm{SL}_n(\mathbb{K})$ is the set of $n \times n$ matrices with determinant 1, and the determinant is a polynomial function on $\mathbb{K}^{n \times n}$. Therefore we can define the polynomial

$$f = \det(A) - 1 \in \mathbb{K}[x_1, \dots, x_{n \times n}]$$

where A is a $n \times n$ matrix, and $\mathrm{SL}_n(\mathbb{K}) = \mathbb{V}(\{f\})$.

(e) **TODO:**

Problem 3. Defining Ideal of an Algebraic Set. Let $V \subset \mathbb{K}^n$ be an algebraic set. The *defining ideal* of V is the ideal in $\mathbb{K}[x_1, \dots, x_n]$ given by

$$\mathbb{I}(V) := \{f \in \mathbb{K}[x_1, \dots, x_n] \mid f(\mathbf{p}) = 0 \text{ for all } \mathbf{p} \in V\}.$$

- (a) Prove that $\mathbb{I}(V)$ is in fact an ideal.
- (b) Prove that $\mathbb{I}(V)$ is radical.
- (c) If $V = \mathbb{V}(\mathcal{S})$ for some subset $\mathcal{S} \subset \mathbb{K}[x_1, \dots, x_n]$, show that $\langle \mathcal{S} \rangle \subset \sqrt{\langle \mathcal{S} \rangle} \subset \mathbb{I}(V)$.
- (d) Let $\mathbb{K} = \mathbb{R}$. Give an example to show that the inclusion $\sqrt{\langle \mathcal{S} \rangle} \subset \mathbb{I}(V)$ can be proper.
- (e) If $V = \mathbb{V}(\mathcal{S})$ for some subset $\mathcal{S} \subset \mathbb{K}[x_1, \dots, x_n]$, show that $\mathbb{V}(\mathcal{S}) = \mathbb{V}(\mathbb{I}(V))$.

Solution.

- (a) Let $g \in \mathbb{K}[x_1, \dots, x_n]$ and $f_1, f_2 \in \mathbb{I}(V)$. That is, $f_1(\mathbf{p}) = f_2(\mathbf{p}) = 0$.

Subgroup Under Addition: Note that,

$$(f_1 + f_2)(\mathbf{p}) = f_1(\mathbf{p}) + f_2(\mathbf{p}) = 0$$

The zero polynomial $0(\mathbf{p}) = 0$ acts as the identity, and

$$f_1(\mathbf{p}) + (-f_1)(\mathbf{p}) = 0(\mathbf{p}) = 0$$

gives us the additive inverse.

Closed Under Multiplication: Notice,

$$(g \cdot f)(\mathbf{p}) = g(\mathbf{p}) \cdot f_1(\mathbf{p}) = g(\mathbf{p}) \cdot 0 = 0$$

Thus $g \cdot f \in \mathbb{I}(V)$ and $\mathbb{I}(V)$ is an ideal.

- (b) We want to show that $\mathbb{I}(V) = \sqrt{\mathbb{I}(V)}$.

Note that the containment $\mathbb{I}(V) \subseteq \sqrt{\mathbb{I}(V)}$ is obvious. To show the opposite containment take $f \in \sqrt{\mathbb{I}(V)}$. Then for some n we have,

$$f^n(\mathbf{p}) = (f(\mathbf{p}))^n = 0$$

But since $\mathbb{K}$ is a domain, this implies that $f(\mathbf{p}) = 0$. Hence $f \in \mathbb{I}(V)$, $\mathbb{I}(V) \supseteq \sqrt{\mathbb{I}(V)}$, and $\mathbb{I}(V) = \sqrt{\mathbb{I}(V)}$.

- (c) Let $V = V(\mathcal{S})$. We know that $\langle \mathcal{S} \rangle \subseteq \sqrt{\langle \mathcal{S} \rangle}$. So showing that $\sqrt{\langle \mathcal{S} \rangle} \subset \mathbb{I}(V)$ suffices. Note that $\mathbb{I}(V) = \mathbb{I}(V(\mathcal{S}))$ which is the set of all functions that evaluate to 0 at $\mathbf{p}$, for all $\mathbf{p} \in V(\mathcal{S})$. And $V(\mathcal{S})$ is the set of all points $\mathbf{p}$ such that all functions $f \in \mathcal{S}$ evaluate to 0 at $\mathbf{p}$. Let $f \in \sqrt{\langle \mathcal{S} \rangle}$. Then by definition for some n , $f^n \in \langle \mathcal{S} \rangle$. We can then write,

$$f^n = \sum_i h_i(\mathbf{p}) g_i(\mathbf{p}), \quad h_i \in \mathbb{K}[x_1, \dots, x_n], g_i \in \mathcal{S}$$

But then,

$$f^n(\mathbf{p}) = \sum_i h_i(\mathbf{p}) g_i(\mathbf{p}) = \sum_i h_i(\mathbf{p}) \cdot 0 = 0$$

Moreover since $\mathbb{K}$ is a domain, $f^n(\mathbf{p}) = 0 \implies f(\mathbf{p}) = 0 \implies f$ is a function that evaluates to 0 at $\mathbf{p}$, hence $f \in \mathbb{I}(V)$.

- (d) Consider the set $\mathcal{S} = \{x^2 + 1\}$. We have,

$$\sqrt{\langle \mathcal{S} \rangle} = \sqrt{\langle x^2 + 1 \rangle}$$

And, because the polynomial has no solutions over $\mathbb{R}$, we have $V = V(\mathcal{S}) = \emptyset$. But $\mathbb{I}(V(\mathcal{S})) = \mathbb{R}[x]$. Also note that since $\langle \mathcal{S} \rangle$ is a prime ideal ($x^2 + 1$ is irreducible), we have $\langle x^2 + 1 \rangle = \sqrt{\langle x^2 + 1 \rangle} \subset \mathbb{R}[x]$. Hence, the containment $\sqrt{\langle \mathcal{S} \rangle} \subset \mathbb{I}(V)$ can be proper.

- (e) Let $V = V(\mathcal{S})$.

$$V(\mathcal{S}) \subseteq V(\mathbb{I}(V(\mathcal{S})))$$

This containment is obvious by construction. $\mathbb{I}(V(\mathcal{S}))$ is the set of all functions that vanish at all points in $V(\mathcal{S})$. And $V(\mathbb{I}(V(\mathcal{S})))$ is the set of all points that all functions in $\mathbb{I}(V(\mathcal{S}))$ vanish at. So, $V(\mathcal{S}) \subseteq V(\mathbb{I}(V(\mathcal{S})))$.

$$V(\mathcal{S}) \supseteq V(\mathbb{I}(V(\mathcal{S})))$$

Note that by definition $\mathcal{S} \subseteq \mathbb{I}(V(\mathcal{S}))$. Then by 2(a) $V(\mathbb{I}(V(\mathcal{S}))) \subseteq V(\mathcal{S})$.

Hence, $V(\mathcal{S}) = V(\mathbb{I}(V(\mathcal{S})))$.

So far you have built two order-reversing operations:

$$\mathcal{S} \longmapsto \mathbb{V}(\mathcal{S}) \quad \text{and} \quad V \longmapsto \mathbb{I}(V).$$

You have shown that passing from equations to a zero set only remembers the radical of the ideal they generate, and Question 4d shows that over a non-algebraically closed field this can lose information. The next theorem says that over an algebraically closed field this is the only ambiguity.

Theorem 0.2 (Hilbert's Nullstellensatz II). *If $\mathbb{K}$ is algebraically closed and $V = \mathbb{V}(\mathcal{S}) \subset \mathbb{K}^n$ is an algebraic set, for some subset $\mathcal{S} \subset \mathbb{K}[x_1, \dots, x_n]$, then $\mathbb{I}(V) = \sqrt{\langle \mathcal{S} \rangle}$.*

At this point the basic affine dictionary is taking shape; Question (4) asks you to make the correspondence between algebraic sets and radical ideals completely explicit. After that we turn to coordinate rings, which package an algebraic set V into algebra and will let us recover from $\mathbb{K}[V]$ its points, its topology, and ultimately its morphisms.

Problem 4. Let $\mathbb{K}$ be algebraically closed. Use Hilbert's Nullstellensatz II (above) to give an explicit inclusion-reversing bijection between algebraic sets in $\mathbb{K}^n$ and radical ideals.

Solution.

Let $\mathcal{S} \subset \mathbb{K}[x_1, \dots, x_n]$.

$$\mathbb{V}(\mathcal{S}) \xrightarrow{\mathbb{I}} \sqrt{\langle \mathcal{S} \rangle}$$

$$\mathbb{V}(\mathcal{S}) \xleftarrow{\mathbb{V}} \sqrt{\langle \mathcal{S} \rangle}$$

TODO: Diagram.

Problem 5. Coordinate Rings.

Let $V \subset \mathbb{K}^n$ be an algebraic set and $\mathcal{R} = \text{Hom}_{\text{Set}}(V, \mathbb{K})$ be the ring of all (set) functions from V to $\mathbb{K}$. In this question we study the restriction map from $\mathbb{K}[x_1, \dots, x_n]$ to $\mathcal{R}$ given by:

$$\begin{array}{ccc} \mathbb{K}[x_1, \dots, x_n] & \xrightarrow{\rho} & \mathcal{R} \\ f & \longmapsto & f|_V \end{array}.$$

Here we are viewing $f \in \mathbb{K}[x_1, \dots, x_n]$ as defining a function $\mathbb{K}^n \rightarrow \mathbb{K}$ given by $\mathbf{p} \mapsto f(\mathbf{p})$. We call the image of ρ the *coordinate ring* of V and denote it $\mathbb{K}[V]$.

- (a) Prove that ρ is in fact a ring homomorphism.
- (b) Verify that each of the following functions lies in the coordinate ring of the given algebraic set.
 - (i) Let $V = \mathbb{V}(x^2 + y^2 - 1) \subset \mathbb{R}^2$ and consider the function $V \rightarrow \mathbb{R}$ given by $(x, y) \mapsto e^{x^2+y^2}$.
 - (ii) Let $V = \mathbb{V}(xy - 1) \subset \mathbb{K}^2$ and consider the function $V \rightarrow \mathbb{K}$ given by $(x, y) \mapsto \frac{1}{x}$.
- (c) Identify the kernel of ρ and use it to give an explicit presentation for $\mathbb{K}[V]$ as a $\mathbb{K}$ -algebra.
- (d) The i -th coordinate of $\mathbb{K}^n$ is the linear functional $e_i^\vee : \mathbb{K}^n \rightarrow \mathbb{K}$ sending $(a_1, \dots, a_n) \mapsto a_i$. Explain why $e_i^\vee \in \mathbb{K}[\mathbb{K}^n]$, and under the presentation given in part (c), which polynomial corresponds to $e_i^\vee$.
- (e) Explain why $\mathbb{K}[V]$ is called the coordinate ring of V . Hint: Think about the previous part.
- (f) Describe the coordinate ring for the following algebraic sets:
 - (i) $\mathbb{V}(y - x^2) \subset \mathbb{K}^2$.
 - (ii) $\mathbb{V}(y^2, xy, x^{17}) \subset \mathbb{K}^2$.
 - (iii) $\mathbb{V}(x^6 - 3x^5y + 3x^4y^2 - x^3y^3) \subset \mathbb{K}^2$.
 - (iv) The set of 3×3 matrices over $\mathbb{F}_5$ of determinant 1 and trace 0. (Hint/Warning: There will likely be 9 more polynomials in the defining ideal than you expect, coming from the fact that finite fields have a special automorphism. This highlights another way we might get a counterexample for Question 4d.)

Solution.

- (a) Let $f_1, f_2 \in \mathbb{K}[x_1, \dots, x_n]$. Note that addition and multiplication is defined pointwise in $\mathbb{R}$. To see if ρ is a ring homomorphism, we check if the three properties hold:

Addition:

$$\rho(f_1) + \rho(f_2)(p) = f_1|_V(p) + f_2|_V(p) = (f_1 + f_2)|_V(p) = \rho(f_1 + f_2)(p)$$

Multiplication:

$$\rho(f_1) \cdot \rho(f_2)(p) = f_1|_V(p) \cdot f_2|_V(p) = (f_1 \cdot f_2)|_V(p) = \rho(f_1 \cdot f_2)(p)$$

Unital:

$$\rho(1)(p) = 1|_V(p) = 1$$

- (b) (i) Let $f : (x, y) \mapsto e^{x^2+y^2}$. Note that on V , $x^2 + y^2 = 1$ then, $f|_V = e^{x^2+y^2} = e^1$. Which behaves like the constant polynomial function $g(x, y) = e$ and hence lies in the ring of coordinates of V .
- (ii) Let $f : (x, y) \mapsto \frac{1}{x}$. Note that on V , $\frac{1}{x} = y$, and $x \neq 0$ because $\mathbb{K}$ is a domain. Then, $f|_V = \frac{1}{x} = y$. Which behaves like the polynomial function $g(x, y) = y$ and hence lies in the ring of coordinates of V .
- (c) The kernel of ρ is $\mathbb{I}(V)$.

$$\mathbb{K}[V] \cong \mathbb{K}[x_1, \dots, x_n]/\mathbb{I}(V)$$

- (d) The coordinate ring of $\mathbb{K}^n$, $\mathbb{K}[\mathbb{K}^n]$ is the ring of all set functions $\mathbb{K}^n \rightarrow \mathbb{K}$.

$$e_i^\vee(x_1, \dots, x_n) = x_i$$

which corresponds to the polynomial function $g(x_1, \dots, x_n) = x_i$. Hence, $e_i^\vee \in \mathbb{K}[\mathbb{K}^n]$. Under the presentation given in part (c) we have,

$$\mathbb{K}[\mathbb{K}^n] \cong \mathbb{K}[x_1, \dots, x_n]/\mathbb{I}(\mathbb{K}^n) = \mathbb{K}[x_1, \dots, x_n]/\{0\} = \mathbb{K}[x_1, \dots, x_n]$$

So, $e_i^\vee$ corresponds to the coset $[x_i] = x_i$.

- (e) $\mathbb{K}[V]$ is the coordinate ring of V because every polynomial is a combination of coordinate functions restricted to V . And $\mathbb{K}[V]$ is generated by these functions on V .

$$\mathbb{K}[V] = \mathbb{K}[e_1^\vee|_V, \dots, e_n^\vee|_V]$$

- (f) (i) Let $V = \mathbb{V}(y - x^2)$.

$$\mathbb{K}[V] \cong \mathbb{K}[x, y]/\mathbb{I}(y - x^2) = \mathbb{K}[x, x^2] = \mathbb{K}[x]$$

(ii) Let $V = \mathbb{V}(y^2, xy, x^{17})$.

Note that $I(V) = \sqrt{\langle y^2, xy, x^{17} \rangle}$.

Since $\mathbb{K}$ is a field, if $(a, b) \in V$ then

$$a^{17} = 0 \implies a = 0$$

and

$$b^2 = 0 \implies b = 0$$

Then, $V = \{(0, 0)\}$. And $I(V) = I(\{(0, 0)\}) = \langle x, y \rangle$.

$$\mathbb{K}[V] \cong \mathbb{K}[x, y]/I(V) = \mathbb{K}[x, y]/\langle x, y \rangle = \mathbb{K}$$

(iii) Let $V = \mathbb{V}(x^6 - 3x^5y + 3x^4y^2 + x^3y^3) = \mathbb{V}(x^3(x - y)^3)$.

Then if $(a, b) \in V$, either $a^3 = 0 \implies a = 0$ or $(x - y)^3 = 0 \implies x = y$.

Then $V = \{(0, a), (a, a) \mid a \in \mathbb{K}\} = \mathbb{V}(x) \cup \mathbb{V}(x - y)$.

$$I(V) = I(\mathbb{V}(x) \cup \mathbb{V}(x - y)) = I(\mathbb{V}(x)) \cap I(\mathbb{V}(x - y)) = \langle x \rangle \cap \langle x - y \rangle = \langle x(x - y) \rangle$$

And

$$\mathbb{K}[V] \cong \mathbb{K}[x, y]/\langle x(x - y) \rangle$$

(iv) Let $A = (x_{ij}) \in \text{SL}_3(\mathbb{F}_5)$, and $V = \mathbb{V}(\det(A) - 1, \text{tr}(A))$. Note that the trace formula is $x_{11} + x_{22} + x_{33}$, and we use fermat's little theorem to enforce staying in $\mathbb{F}_5$. Then the coordinate ring is,

$$\mathbb{K}[V] = \mathbb{F}_5[x_{11}, \dots, x_{33}]/\langle \det(A) - 1, x_{11} + x_{22} + x_{33}, x_{ij}^5 - x_{ij} \rangle$$

You should now think of $\mathbb{K}[V]$ as the algebraic avatar of V : its elements are precisely the polynomial functions on V , and the coordinate functions on $\mathbb{K}^n$ descend to generators of $\mathbb{K}[V]$. The next step is to recognize points of V purely algebraically, either as $\mathbb{K}$ -algebra maps $\mathbb{K}[V] \rightarrow \mathbb{K}$ or as maximal ideals whose residue fields are isomorphic to $\mathbb{K}$ as $\mathbb{K}$ -algebras.

Problem 6. Let R be a finitely generated $\mathbb{K}$ -algebra. Fix a presentation $R \cong \mathbb{K}[x_1, \dots, x_n]/I$. Describe a natural bijection between the following three sets:

- (i) points in $V(I) \subset \mathbb{K}^n$,
- (ii) maximal ideals in R with residue field isomorphic to $\mathbb{K}$ as a $\mathbb{K}$ -algebra, and
- (iii) elements of $\text{Hom}_{\mathbb{K}\text{-alg}}(R, \mathbb{K})$, the set of morphisms from R to $\mathbb{K}$ in the category of $\mathbb{K}$ -algebras.

Solution.

Let $\mathbb{K}$ be algebraically closed.

Claim: Let $\pi: \mathbb{K}[x_1, \dots, x_n] \rightarrow \mathbb{K}[x_1, \dots, x_n]/I = R$, and let $J \subset R$ be maximal. We claim that $\pi^{-1}(J)$ is maximal in $\mathbb{K}[x_1, \dots, x_n]$.

Proof. We know that $\pi^{-1}(J)$ is an ideal. Let M be a maximal ideal containing $\pi^{-1}(J)$. Since π is surjective, we know that $\pi(M)$ is an ideal (worksheet 1.1, Q4.4), and $J \subseteq \pi(M)$. Then, since J is maximal, either $\pi(M) = J$ or $\pi(M)$ is the whole ring. $\pi(M) \neq R$ since ring homomorphisms are unital and $1 \notin M$. Hence, $M = \pi^{-1}(J)$ or $\pi^{-1}(J)$ is maximal. $\square$

(i) $\iff$ (ii)

Recall that $V(I)$ contains points $\mathbf{p} = (a_1, \dots, a_n)$ such that $f(\mathbf{p}) = 0$ for all $f \in I$. Also,

$$\text{ev}_{\mathbf{p}}: \mathbb{K}^n \rightarrow \mathbb{K}$$

By Hilbert's Nullstellensatz I, we know that maximal ideals of $\mathbb{K}[x_1, \dots, x_n]$ are of the form $\langle x_1 - a_1, \dots, x_n - a_n \rangle$, for some point $(a_1, \dots, a_n)$.

Problem 7. Zariski Topology on Algebraic Sets. Let $V \subseteq \mathbb{K}^n$ be an algebraic set. Define the *Zariski topology* on V by declaring the closed sets to be the subsets of V of the form $V(I)$ for some ideal $I \subset \mathbb{K}[x_1, \dots, x_n]$. In other words, the closed sets are precisely the *algebraic subsets* of V .

- (a) Verify that this definition indeed determines a topology on V .

- (b) For $f \in \mathbb{K}[V]$, define

$$D(f) = \{p \in V \mid f(p) \neq 0\}.$$

Show that $\{D(f)\}$ form a basis for the Zariski topology on V .

- (c) For each point $p \in V$, let

$$\mathfrak{m}_p = \{f \in \mathbb{K}[V] \mid f(p) = 0\}$$

be the corresponding maximal ideal. Show that the map

$$\begin{aligned} V &\longrightarrow \operatorname{Spec}(\mathbb{K}[V]) \\ p &\longmapsto \mathfrak{m}_p \end{aligned}$$

is a homeomorphism onto its image, which consists of those maximal ideals whose residue field is isomorphic to $\mathbb{K}$ as a $\mathbb{K}$ -algebra.

- (d) If $\mathbb{K}$ is algebraically closed how can the preceding part be rephrased in terms of $\operatorname{mSpec}(\mathbb{K}[V])$?
- (e) Is the Zariski topology on V Hausdorff in general? If not, when is it?

Solution.

- (a) $V = \mathbb{V}(I)$ for some $I \subset \mathbb{K}[x_1, \dots, x_n]$. The closed sets of V are of the form $\mathbb{V}(J)$, where $I \subset J \subset \mathbb{K}[x_1, \dots, x_n]$. We need to check the following:

- (i) $\emptyset$ and V are closed.

Note $\mathbb{V}(I) = V$, and $\mathbb{V}(\mathbb{K}[x_1, \dots, x_n]) = \emptyset$ is closed.

- (ii) Finite union of closed sets is closed.

Let $\mathbb{V}(J), \mathbb{V}(K) \subseteq V$ be subsets. Then, $\mathbb{V}(I) \cup \mathbb{V}(J)$ is the set of points in $\mathbb{K}^n$ such that for all $p \in \mathbb{V}(I) \cup \mathbb{V}(J)$ either $f(p) = 0$ or $g(p) = 0$.

- (iii) Arbitrary intersection of closed sets is closed.

By now you have recovered from $\mathbb{K}[V]$ not just the points of V , but also its Zariski topology. The next goal is to understand maps in the same way: a morphism of algebraic sets should be encoded contravariantly by a $\mathbb{K}$ -algebra homomorphism of coordinate rings.

Problem 8. Morphisms of Algebraic Sets. A *polynomial map* from $\mathbb{K}^n$ to $\mathbb{K}^m$ is a function of the form:

$$\Phi : \mathbb{K}^n \longrightarrow \mathbb{K}^m \quad \mathbf{p} \mapsto (f_1(\mathbf{p}), \dots, f_m(\mathbf{p})).$$

where $f_1, \dots, f_m \in \mathbb{K}[x_1, \dots, x_n]$. A *morphism of algebraic sets* $V \rightarrow W$ is defined to be the restriction of such a polynomial map Φ with $\Phi(V) \subset W$. Fix a polynomial map Φ as above.

- (a) Show that there is an induced $\mathbb{K}$ -algebra homomorphism

$$\phi : \mathbb{K}[\mathbb{K}^m] \longrightarrow \mathbb{K}[\mathbb{K}^n] \quad \text{given by} \quad g \mapsto g \circ \Phi.$$

Describe ϕ explicitly in terms of the coordinate function generators of $\mathbb{K}[\mathbb{K}^m]$.

- (b) Let $V \subset \mathbb{K}^n$ and $W \subset \mathbb{K}^m$ be algebraic sets. Show that $\Phi(V) \subset W$ if and only if $\phi(\mathbb{I}(W)) \subset \mathbb{I}(V)$.
- (c) Deduce that Φ restricts to a morphism of algebraic sets $\Phi|_V : V \longrightarrow W$ if and only if the composition

$$\mathbb{K}[\mathbb{K}^m] \xrightarrow{\phi} \mathbb{K}[\mathbb{K}^n] \xrightarrow{\rho} \mathbb{K}[V]$$

factors through $\mathbb{K}[W]$. Here ρ is the restriction map.

- (d) For each of the given maps of coordinate rings, describe the corresponding map of algebraic sets. When possible draw pictures. Compare to the corresponding maps on spectra.
- (i) $\mathcal{C}[x, y] \rightarrow \mathcal{C}[t]$ given by $x \mapsto t$ and $y \mapsto t$.
 - (ii) $\mathcal{C}[t] \rightarrow \mathcal{C}[x, y]$ given by $t \mapsto x$.
 - (iii) $\mathcal{C}[t] \rightarrow \mathcal{C}[x, y, t]/\langle xy - t \rangle$ given by $t \mapsto t$.

Problem 9. Prove that a morphism of algebraic sets is continuous with respect to the Zariski topology.

Problem 10. Show that the image of $\mathbb{K} \rightarrow \mathbb{K}^3$ sending $t \rightarrow (t, t^2, t^3)$ is an algebraic set defined by two polynomials. Verify your answer by constructing the corresponding ring map in *Macaulay2*.

Problem 11. Prove or disprove: The image of every polynomial map $\mathbb{K}^n \rightarrow \mathbb{K}^m$ is an algebraic set. (Warning: this might be field dependent. As a hint, if $\mathbb{K}$ is finite, which subsets of $\mathbb{K}^n$ are algebraic sets?)

You now have the main algebra-geometry dictionary in hand. Over any field $\mathbb{K}$, every algebraic set $V \subset \mathbb{K}^n$ has a coordinate ring $\mathbb{K}[V]$, points of V are the same as $\mathbb{K}$ -algebra maps $\mathbb{K}[V] \rightarrow \mathbb{K}$, and morphisms $V \rightarrow W$ correspond contravariantly to $\mathbb{K}$ -algebra homomorphisms $\mathbb{K}[W] \rightarrow \mathbb{K}[V]$. Thus the assignment

$$V \longmapsto \mathbb{K}[V]$$

always gives a contravariant bridge from geometry to algebra. When $\mathbb{K}$ is algebraically closed, Hilbert's Nullstellensatz says that every finitely generated reduced $\mathbb{K}$ -algebra comes from an algebraic set, so this bridge becomes an anti-equivalence of categories.

Problem 12. Putting It All Together. Assume that $\mathbb{K}$ is algebraically closed. Use the previous parts together with Hilbert's Nullstellensatz II to show that every finitely generated reduced $\mathbb{K}$ -algebra is isomorphic to $\mathbb{K}[V]$ for some algebraic set V . Conclude that the assignment $V \mapsto \mathbb{K}[V]$ defines a (anti-) equivalence from the category of algebraic sets over $\mathbb{K}$ to the category of finitely generated reduced $\mathbb{K}$ -algebras.